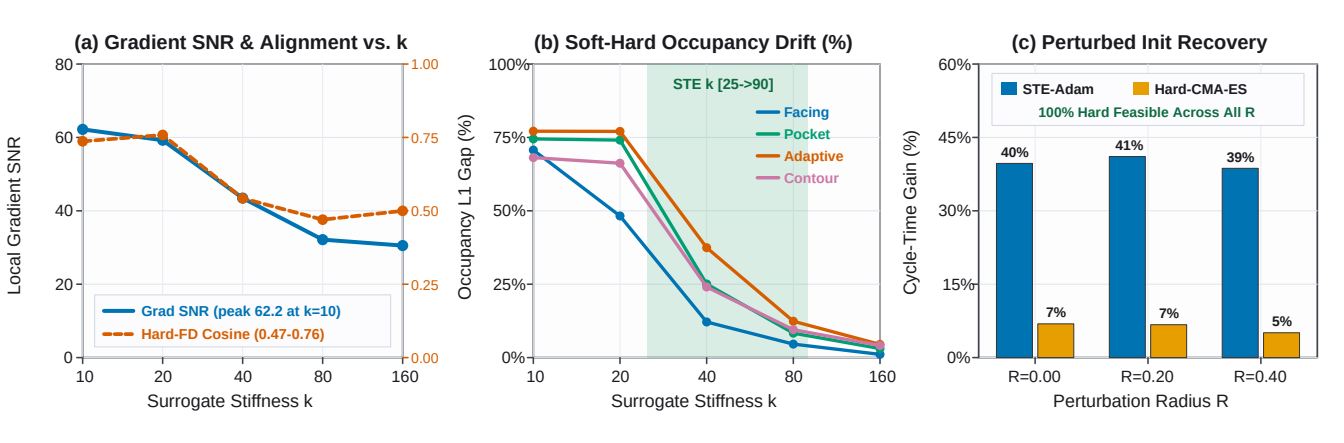
(a) Gradient SNR & Alignment vs. k
Local Gradient SNR
Surrogate Stiffness k
Grad SNR (peak 62.2 at k=10)
Hard-FD Cosine (0.47-0.76)
(b) Soft-Hard Occupancy Drift (%)
Occupancy L1 Gap (%)
Surrogate Stiffness k
STE k [25->90]
Facing
Pocket
Adaptive
Contour
(c) Perturbed Init Recovery
Cycle-Time Gain (%)
Perturbation Radius R
STE-Adam
Hard-CMA-ES
100% Hard Feasible Across All R
40%
7%
41%
7%
39%
5%
R=0.00
R=0.20
R=0.40